

Gene Expression Classification with Support Vector Machines

Statistics for Data Science — Homework #3

University of Trento

MSc in Data Science

Author: Henri Vasserot

Date: May 2026

Objective

This report evaluates Support Vector Machine classifiers for leukemia subtype prediction using high-dimensional gene expression data. Predictive performance is compared before and after variance-based dimensionality reduction.

The report is fully reproducible and the complete source code is available through the accompanying GitHub repository.

GitHub repository



Complete reproducible code and notebook:
https://github.com/lolipop913/homework03_svm_gene_expression.git

Gene Expression Classification with Support Vector Machines

This study investigates the use of Support Vector Machine (SVM) classifiers for predicting leukemia subtypes from high-dimensional gene expression data. The dataset consists of expression measurements for 2000 genes collected from 79 patients belonging to two groups: patients with a chromosomal translocation and cytogenetically normal patients.

The analysis is conducted in a setting where the number of predictors greatly exceeds the number of observations, increasing the risk of overfitting and unstable predictive performance. In such contexts, model evaluation and validation procedures become particularly important in order to distinguish reproducible patterns from noise present in the training data.

Two SVM variants are compared: a linear kernel and a radial basis function (RBF) kernel. Predictive performance is evaluated using cross-validation together with an independent test set. In addition, a variance-based gene filtering strategy is investigated in order to assess whether restricting the analysis to the most variable genes is associated with improved predictive performance and reduced overfitting.

1. Imports and setup

2. Load data

The dataset contains gene expression measurements for 79 patients and 2000 genes together with a binary response variable indicating the leukemia subtype. This corresponds to a high-dimensional setting in which the number of predictors greatly exceeds the number of observations, increasing the risk of overfitting and unstable predictive performance.

Before fitting the models, we inspect the dimensionality of the dataset and the class distribution in order to assess potential class imbalance and to characterize the structure of the prediction problem.

```
X shape: (79, 2000)
y shape: (79,)
Class distribution:
-1    42
 1    37
Name: count, dtype: int64
```

The dataset contains 79 observations and 2000 gene expression variables, corresponding to a setting where the number of predictors greatly exceeds the number of samples. Such configurations increase the risk of overfitting and may lead to unstable predictive performance across different training subsets.

The two leukemia subgroups are relatively balanced (42 versus 37 observations), reducing the risk of strong classification bias toward one class. Consequently, overall accuracy remains a reasonably informative evaluation metric, although class-specific performance metrics are also examined later in the analysis.

3. Train/Test split

The dataset is divided into training and test sets in order to evaluate predictive performance on unseen observations. A stratified split is used to preserve class proportions across the two subsets.

Model selection and hyperparameter tuning are performed exclusively on the training set, while the test set is reserved for the final evaluation of generalization performance. This separation is particularly important in high-dimensional settings, where flexible models may otherwise produce overly optimistic performance estimates on the training data.

```
Train size: (55, 2000)
Test size: (24, 2000)
Train class distribution:
-1    0.527273
 1    0.472727
Name: proportion, dtype: float64
Test class distribution:
-1    0.541667
 1    0.458333
Name: proportion, dtype: float64
```

The dataset is divided into 55 training observations and 24 test observations. The class proportions remain similar across the two subsets, confirming that the stratified sampling strategy successfully preserved the balance between leukemia subgroups.

4. Analysis plan

Two Support Vector Machine classifiers are considered: a linear kernel SVM and a radial basis function (RBF) SVM. For both models, the regularization parameter (C) is tuned, while the RBF kernel additionally requires tuning of the parameter (γ), which controls the flexibility of the decision boundary.

Model performance is evaluated using cross-validation on the training set together with a final assessment on an independent test set. Hyperparameter tuning is performed using `GridSearchCV`, and predictive performance is compared using cross-validation accuracy, test accuracy, confusion matrices, and class-specific classification metrics.

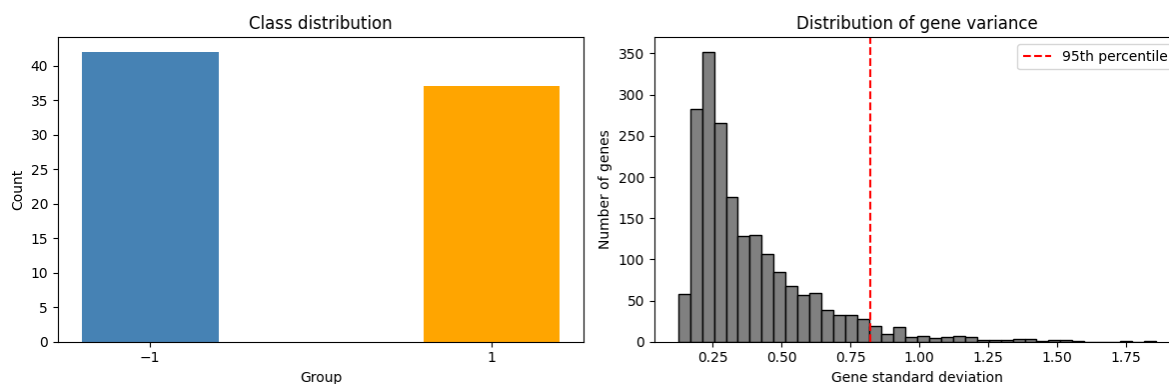
Two preprocessing strategies are investigated. The first uses the complete set of genes, while the second restricts the analysis to the 5% most variable genes in order to reduce dimensionality and limit the influence of weakly informative predictors.

5. Data exploration

We first examine the class distribution together with the variability of gene expression across samples. The two leukemia subgroups appear reasonably balanced, reducing the risk of strong classification bias toward one class.

The distribution of gene standard deviations reveals substantial heterogeneity in expression variability across the dataset. While a relatively small subset of genes exhibits high variability across patients, many genes display limited variation. Low-variance genes are less likely to contribute meaningful discriminative information and may unnecessarily increase the dimensionality of the predictor space.

This observation motivates the use of variance-based filtering, where only the top 5% most variable genes are retained for downstream analysis. Importantly, the filtering procedure is incorporated directly within the machine learning pipeline, ensuring that feature selection is performed independently within each resampling step and thereby avoiding information leakage.



Total number of genes: 2000

Top 5% genes: 100

The class distribution confirms that the two leukemia subgroups are relatively balanced, with 42 observations in class (-1) and 37 observations in class (1). Consequently, overall accuracy remains a reasonably informative evaluation metric for model comparison.

The distribution of gene standard deviations reveals substantial heterogeneity in expression variability across the dataset. While most genes exhibit limited variation across patients, a smaller subset displays markedly larger standard deviations. This pattern supports the use of variance-based filtering, since highly variable genes are more likely to contain discriminative information.

Using the 95th percentile threshold retains 100 genes out of the original 2000 predictors. This substantially reduces the dimensionality of the predictor space while preserving the most variable expression patterns for downstream classification.

6. Pipelines and utility functions

Two machine learning pipelines are defined: one using the complete set of genes and another incorporating variance-based gene filtering. In both cases, preprocessing and model fitting are integrated within a single pipeline structure to ensure that all transformations are applied consistently during cross-validation.

Embedding the filtering procedure directly within the pipeline prevents information leakage by ensuring that feature selection is performed independently within each resampling step. The same cross-validation strategy is used across all models in order to maintain a consistent comparison between preprocessing approaches and SVM variants.

7. SVM on all features

We first fit Support Vector Machine classifiers using the complete set of available genes in order to establish a baseline predictive performance without prior dimensionality reduction.

Both linear and RBF kernels are evaluated after hyperparameter tuning through cross-validation on the training set. The comparison between kernels allows us to assess whether increased model flexibility is associated with improved predictive performance in this high-dimensional setting.

Because the number of predictors substantially exceeds the number of observations, strong training performance may not necessarily translate into stable generalization performance on unseen data. Particular attention is therefore given to the comparison between cross-validation, training, and test accuracies.

Full features - kernel: rbf
 Full features - C: 10
 Full features - CV accuracy: 0.691
 Full features - train accuracy: 1.0
 Full features - test accuracy: 0.667

For the full-feature analysis, the best-performing configuration selected by cross-validation is an RBF SVM with ($C = 10$) and ($=$) "scale". The relatively large value of (C) indicates a lower tolerance for classification errors on the training data and suggests a comparatively complex decision boundary.

The model achieves a cross-validation accuracy of approximately 0.69 and a test accuracy of 0.67, while the training accuracy reaches 1.00. The substantial gap between training and test performance indicates strong overfitting and suggests that part of the learned structure does not generalize well to unseen observations.

The comparison between kernels further shows that the RBF models outperform the linear models in terms of average cross-validation accuracy, indicating that non-linear decision boundaries may better capture the structure of the dataset. However, the relatively large variability across folds highlights the instability expected in a small high-dimensional dataset.

	kernel	C	gamma	cv_mean	cv_std
13	rbf	100.0	scale	0.691	0.093
10	rbf	10.0	scale	0.691	0.093
0	linear	0.1	NaN	0.673	0.123
1	linear	1.0	NaN	0.673	0.123
7	rbf	1.0	scale	0.673	0.159
3	linear	100.0	NaN	0.673	0.123

The comparison of hyperparameter configurations indicates that the best-performing models are obtained with the RBF kernel. Linear SVM models generally achieve slightly lower cross-validation accuracy, suggesting that non-linear decision boundaries may better capture the structure of the gene expression data.

However, differences between configurations remain moderate, and the observed variability across folds indicates that predictive performance is sensitive to the specific training subsets used during resampling.

	precision	recall	f1-score	support
-1	0.67	0.77	0.71	13
1	0.67	0.55	0.60	11

accuracy			0.67	24
macro avg	0.67	0.66	0.66	24
weighted avg	0.67	0.67	0.66	24

The classification report shows moderately balanced predictive performance across the two leukemia subgroups. Class (-1) achieves slightly higher recall than class (1), indicating that the model identifies observations from the first subgroup more reliably. Overall predictive performance remains moderate, with an accuracy of approximately 0.67 on the test set.

8. SVM with top 5% variance filtering

We next repeat the analysis after restricting the predictors to the 5% most variable genes. This filtering strategy substantially reduces the dimensionality of the predictor space while retaining the genes that exhibit the strongest variation across patients.

The filtering procedure is integrated directly within the machine learning pipeline, ensuring that gene selection is performed independently within each resampling step using only the training data. This prevents information leakage and allows predictive performance to be evaluated consistently across cross-validation folds and on the independent test set.

The filtered analysis is then compared with the full-feature analysis in order to assess whether restricting the model to highly variable genes is associated with improved predictive performance and reduced overfitting.

```
Filtered features - kernel: rbf
Filtered features - C: 1
Filtered features - CV accuracy: 0.782
Filtered features - train accuracy: 0.982
Filtered features - test accuracy: 0.75
Number of selected features: 100
```

After variance-based filtering, the best-performing configuration remains an RBF SVM, although with a smaller regularization parameter ($C = 1$). Compared with the full-feature analysis, the filtered model achieves higher cross-validation accuracy (0.78 versus 0.69) together with improved test accuracy (0.75 versus 0.67).

The training accuracy remains high (0.98), but the gap between training and test performance is smaller than in the full-feature setting. This suggests that restricting the analysis to the most variable genes is associated with reduced overfitting and improved generalization performance.

The comparison between kernels again indicates that the RBF models outperform the linear models, although variability across cross-validation folds remains non-negligible. This indicates that predictive performance is still sensitive to sampling fluctuations despite the reduction in dimensionality.

Only 100 genes are retained after filtering, representing a substantial reduction relative to the original 2000 predictors. These results are consistent with the idea that a large proportion of low-variance genes contributes limited discriminative information in this classification task.

	kernel	C	gamma	cv_mean	cv_std
7	rbf	1.0	scale	0.782	0.093
0	linear	0.1	NaN	0.745	0.134
2	linear	10.0	NaN	0.727	0.115
1	linear	1.0	NaN	0.727	0.115

The filtered analysis again selects an RBF kernel as the best-performing configuration. Compared with the full-feature analysis, the filtered models achieve higher average cross-validation accuracy together with lower variability across folds. This suggests that restricting the analysis to highly variable genes improves the stability of predictive performance.

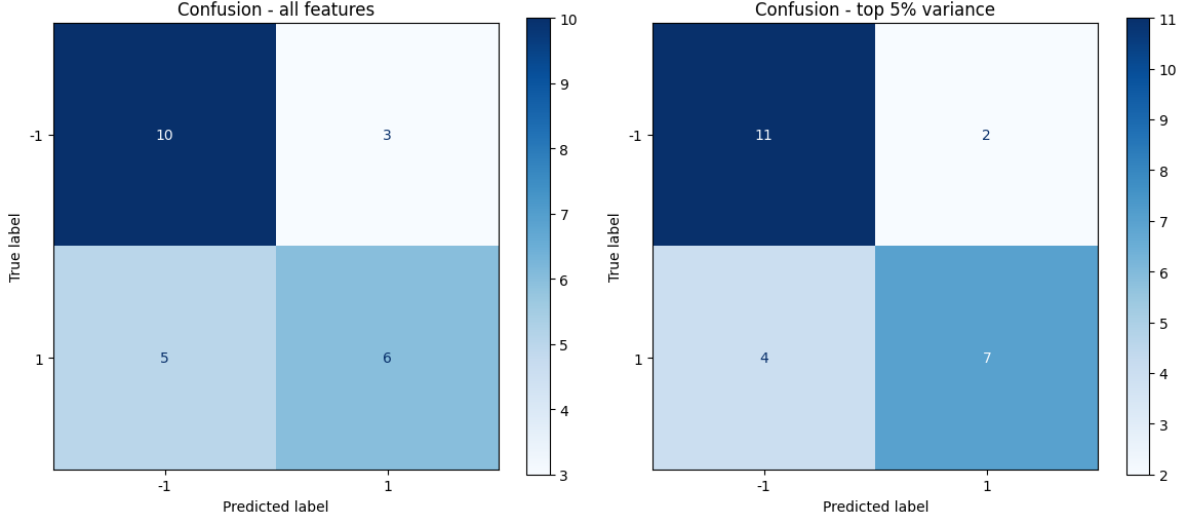
	precision	recall	f1-score	support
-1	0.73	0.85	0.79	13
1	0.78	0.64	0.70	11
accuracy			0.75	24
macro avg	0.76	0.74	0.74	24
weighted avg	0.75	0.75	0.75	24

The classification report indicates a more balanced predictive performance than in the full-feature setting. Recall improves substantially for class (-1), while precision remains relatively similar across the two leukemia subgroups. Overall accuracy increases to approximately 0.75, consistent with the improvements observed in cross-validation performance.

9. Performance comparison

The predictive performance of the different SVM models is compared using identical training and test partitions in order to ensure a consistent evaluation across preprocessing strategies and kernel choices.

Particular attention is given to the effect of variance-based filtering on cross-validation accuracy, test accuracy, training-test performance gaps, and variability across resampling folds. These comparisons allow us to assess whether restricting the analysis to highly variable genes is associated with improved generalization performance and more stable predictive behavior.



The comparative results indicate that variance-based filtering is associated with improved predictive performance across all evaluation stages. The filtered model achieves higher cross-validation accuracy (0.78 versus 0.69) together with higher test accuracy (0.75 versus 0.67) than the model trained on the complete set of genes.

The best-performing configuration in both settings is an RBF SVM. However, the filtered analysis selects a smaller regularization parameter ($(C = 1)$) compared with the full-feature model ($(C = 10)$). This suggests that, after dimensionality reduction, a less complex decision boundary is sufficient to achieve stronger predictive performance.

The difference between training and test accuracy also decreases from 0.33 in the full-feature analysis to 0.23 after filtering. Although both models still exhibit some degree of overfitting, the smaller train-test gap suggests improved generalization to unseen observations.

The confusion matrices further support this interpretation. In the full-feature setting, the classifier produces several misclassifications in both classes, particularly for observations belonging to class (1). After variance filtering, the number of correctly classified observations increases for both groups, resulting in fewer overall classification errors and improved class balance.

Only 100 genes are retained after filtering, representing a substantial reduction relative to the original 2000 predictors. Overall, the results suggest that restricting the analysis to highly variable genes helps concentrate the model on the most informative expression patterns while reducing the influence of weakly discriminative variables.

10. Conclusion

This analysis investigated the use of Support Vector Machine classifiers for leukemia subtype prediction using high-dimensional gene expression data. Both linear and RBF kernels were evaluated on the complete feature space and after variance-based dimensionality reduction.

Across all evaluation stages, the filtered analysis achieved stronger predictive performance than the full-feature approach, including higher cross-validation accuracy, higher test accuracy, smaller training-test performance gaps, and lower variability across resampling folds. The results therefore suggest that restricting the analysis to highly variable genes is associated with improved generalization performance in this dataset.

More broadly, the analysis illustrates the methodological challenges associated with settings in which the number of predictors greatly exceeds the number of observations. In such contexts, highly flexible models may adapt to noise present in the training data, making careful validation procedures and appropriate dimensionality reduction particularly important.

Several limitations should nevertheless be acknowledged. The dataset remains relatively small, the variance threshold is heuristic, and no independent external validation dataset is available. Consequently, the reported performances should be interpreted cautiously and may not fully generalize to independent patient populations.

Overall, the results illustrate how dimensionality reduction combined with regularized machine learning methods can help improve predictive performance in high-dimensional gene expression studies while limiting overfitting.

Final summary

- Full features test accuracy: 0.667
- Filtered features test accuracy: 0.75
- Filtered model train-test gap: 0.232

The filtered model still exhibits some degree of overfitting, although the train-test gap is smaller than in the full-feature analysis.